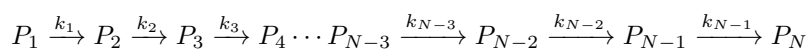


99.500 Applied Mathematics for Engineering

Homework 4 (Q1 is compulsory, followed by either Q2 or Q3)

December 28, 2025

1. In a cutting-edge biotechnology lab, a specialized system is designed to produce a high-value therapeutic protein. This system involves a series of cellular processes in which raw biological materials undergo irreversible transformations to produce the final therapeutic product. The process occurs in $N - 1$ stages, each representing a different step in the cellular machinery, as shown below:



The production starts with an initial substrate, such as a nutrient or precursor protein, denoted as P_1 , which is introduced into the first stage of the cellular process. Within this stage, a reaction converts some of P_1 into an intermediate protein P_2 with rate constant k_1 . This intermediate then proceeds to the next stage, where it is further converted into P_3 with rate constant k_2 . The process continues through a series of stages until the final therapeutic protein P_N is produced with rate constant k_{N-1} . The concentration of P_1 is C_{initial} at time $t = 0$, while all the other proteins have 0 concentration.

The researchers are particularly interested in monitoring the concentration of each protein intermediate throughout the production process to ensure efficiency and optimize yields. To achieve this, use Laplace transform to derive a general formula for the concentration of each protein at any given time, and provide the formula for the concentrations of the first three proteins.

2. Use the separation of variables to find the solution $T(r, z)$ for the following problem:

$$\frac{\partial^2 T}{\partial z^2} + \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial T}{\partial r} \right) = 0 \quad (1)$$

$$T(r, 0) = 0 \quad (2)$$

$$\frac{\partial T}{\partial z} \Big|_{z=\alpha} = 0 \quad (3)$$

$$\frac{\partial T}{\partial r} \Big|_{r=0} = 0 \quad (4)$$

$$\frac{\partial T}{\partial r} \Big|_{r=1} = f(z) \quad (5)$$

where α is constant. What is $T(r, z)$ for the case $f(z) = 1$?

3. Use the separation of variables to find the solution $T(r, z)$ for the following problem:

$$\frac{\partial^2 T}{\partial z^2} + \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial T}{\partial r} \right) = 0 \quad (6)$$

$$T(r, 0) = T_0 \quad (7)$$

$$\frac{\partial T}{\partial z} \Big|_{z=L} = \begin{cases} \beta & r \leq r_1 \\ 0 & r_1 < r < r_2 \end{cases} \quad (8)$$

$$\frac{\partial T}{\partial r} \Big|_{r=0} = 0 \quad (9)$$

$$\frac{\partial T}{\partial r} \Big|_{r=r_2} = 0 \quad (10)$$

where β , r_1 and r_2 are constants.